

$MP = 128 = 2^7 = 7 \text{ bit}$

$MC = 16 = 2^4 = 4 \text{ bit}$

$SimB = 2^2 = 2 \text{ bit}$

$\frac{2^7 - 2^4 - 1}{4} = 1 \quad 2$

$\frac{1}{10 \cdot 625} = 1,6 \text{ ns}$

ADD: \$20, \$20, -12

SW \$0, 0(\$20)

SW \$0, 4(\$20)

SW \$7, 8(\$20)

MOVE \$0, \$0 $\$0 \leftarrow 100$

BEQ \$0, \$5, EXIT

MUL \$0, \$0, 1

ADD \$0, \$0, \$4 $VL:]$

LW \$0, 0(\$0)

ADD \$7, \$0, \$6

i LOOP

LW \$0, 0(\$20)

LW \$0, 4(\$20)

LW \$7, 8(\$20)

ADD: \$20, \$20, \$12

JR \$31

X	52	45	1	44	25	37
B.F.	13	-1	0	11	6	9
S.I.	1	1	0	1	0	1
Hit				X		
SET 0	0	0	0	0	6	6
SET 1	13	11	11	11	11	11

$MOD \frac{B.F.}{2}$

$2^9 \cdot 2^{10} = 18$ BIT MP
 TDA 14 B | 5 BIT OFFSET COMPL ASSO
 MC = 16 BIT MC
 TDA 6 B | S.I. 8 B / OFF 5 BIT N VIEE
 DIMB = 5 BIT
 TDA 3 B / C.I. 71 B / OFF 5 BIT DIRETO

```

move $2, $0      i = 0
beq $2, $5, ESCI  j == N
mului $3, $2, 4
add $3, $4, $3
lw $3, 0($3)
bne $6, $3, ESCI
addui $7, $0, 1
j loop
ESCI: ...
  
```

S ₁	S ₀	F	A	B	C _w
0	0	A'	A'	-1	1
0	1	A+B	A	B'	1
1	0	B-A	A'	B	1
1	1	B'	-1	B'	1

$2^6 \cdot 2^{10} = 2^{16} = 16$ BIT

ASSOCIATIVO COMPL. → TDA 13 B / OFF 3 B
 2 VIEE → TDA 12 B / S.I. 1 B / OFF 3 B
 DIRETO → TDA 11 B / C.I. 2 B / OFF 3 B

```

$A = V[0]
$5 = N
$7 = MAX
$2 = i

addi $23, $23, -12
sw $5, 0($23)
sw $3, 4($23)
sw $10, 8($23)
mului $8, $0, 1      i = 4
lw $7, 0($4)         MAX = V[0]

FOR: beq $8, $3, EXIT
mului $3, $8, 4      i = 4
add $3, $3, $4
lw $3, 0($3)
addi $8, $8, 1
slt $10, $7, $3      if V[i] > MAX
beq $10, $0, EXIT
move $7, $3
j FOR

EXIT: lw $5, 0($23)
      lw $3, 4($23)
      lw $10, 8($23)
      addi $23, $23, 12
      jr $31
  
```

ESERCIZIO 3 (7 punti)

Spiegare le seguenti istruzioni Assembly MIPS ed il loro effetto sui registri e sull'area di memoria rappresentati a fianco:

```

addi $10, $0, 20
move $5, $0
move $6, $0
c: lw $4, A($5)
   add $6, $6, $4
   addi $5, $5, 4
   bne $5, $10, c
   sw $6, A($5)
  
```

Indirizzo registro	Contenuto	Indirizzo memoria	Contenuto
\$4	10 -7	A	10
\$5	0 4 20	A+4	-2
\$6	0 10 11	A+8	5
\$10	20 20	A+12	5
		A+16	-7
		A+20	11

2⁰ 2¹ 2² 2³ BIT DI CONTROLLO

1	2	3	4	5	6	7	8	9	10	11	12
C ₀	C ₁	B ₀	C ₂	B ₁	B ₂	B ₃	C ₃	B ₄	B ₅	B ₆	B ₇
0	1	0	1	0	0	0	1	0	0	0	1

● 2⁰ BIT DI INFORMAZIONE → M₁, M₂, ..., M₈

● 4 BIT DI CONTROLLO

● LA STRINGA FINALE, BIT DI CONTROLLO + BIT DI INFORMAZIONE, È DI 12 BIT

● I BIT DI CONTROLLO STRAORDINARI NELLE POSIZIONI 2⁰, 2¹, 2², 2³, ..., 2ⁿ

DOBBIAMO RITROVARE IL VALORE ASSOCIATO A CIASCUN BIT DI CONTROLLO, DIPENDE DAL CODICE DA RAPPRESENTARE

I BIT DI CONTROLLO SONO CALCOLATI COME (XOR O EXOR)

SE C'È UN NUMERO PARI DI BIT A 1, IL RISULTATO DELLA XOR È ZERO
SE C'È UN NUMERO DISPAREGLIA DI BIT A 1, IL RISULTATO DELLA XOR È 1

C₀ = XOR(M₁, M₂, M₃, M₄) = 0

C₀ = 0 → C₀' = 0

E₀ = XOR(C₀, C₀') = XOR(0, 0) = 0

C₁ = XOR(M₁, M₃, M₅, M₇) = 0

C₁ = 0 → C₁' = 0

E₁ = XOR(C₁, C₁') = XOR(0, 0) = 0

C₂ = XOR(M₂, M₃, M₆, M₇) = 1

C₂ = 1 → C₂' = 1

E₂ = XOR(C₂, C₂') = XOR(1, 1) = 0

C₃ = XOR(M₃, M₆, M₇, M₈) = 1

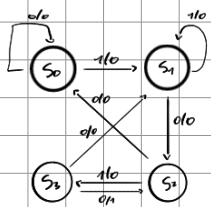
C₃ = 1 → C₃' = 1

E₃ = XOR(C₃, C₃') = XOR(1, 1) = 0

IN GENERALE, IL BIT IN POSIZIONE n È "CONTROLLATO" DA QUEI BIT DI CONTROLLO C_i TALI CHE → $\sum C_i 2^i = n$

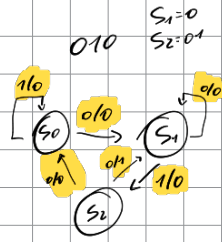
BIT POSITION	C ₀	C ₁	C ₂	C ₃	M _i
1	0	0	0	1	C ₀ = 0
2	0	0	1	0	C ₁ = 1
3	0	0	1	1	M ₁ = 0
4	0	1	0	0	C ₂ = 1
5	0	1	0	1	M ₂ = 0
6	0	1	1	0	M ₃ = 0
7	0	1	1	1	M ₄ = 0
8	1	0	0	0	C ₃ = 1
9	1	0	0	1	M ₅ = 0
10	1	0	1	0	M ₆ = 0
11	1	0	1	1	M ₇ = 0
12	1	1	0	0	M ₈ = 1

1010



S₁ = 1
S₂ = 10
S₃ = 101
S₄ = 1010

1010



S₁ = 0
S₂ = 01

\$4 = A
\$5 = B
\$6 = i

ADDI \$20, \$25, -12
SW \$7, 0(\$25)
SW \$8, 4(\$25)
SW \$31, 8(\$25)
MUL \$7, \$4, \$4
MUL \$8, \$5, \$5
ADDI \$7, \$7, \$8
JAL SORT
LW \$7, 0(\$25)
LW \$8, 4(\$25)
LW \$31, 8(\$25)
ADDI \$25, \$25, 12
JR \$31

\$9V
\$8V

$$\left. \begin{array}{l} (2^{25} - 1) + (1 - 2^{-10}) \text{ MAX} \\ 2^{-10} \text{ MIN} \end{array} \right\} \text{VIRGOLA FISSA}$$

$$\left. \begin{array}{l} (2 - 2^{-23}) \cdot 2^{-13} = (2^{10} - 2^{-100}) \text{ MAX} \\ 1 \cdot 2^{-106} \text{ MIN} \end{array} \right\} \text{VIRGOLA MOBILE}$$

$$0,79 \cdot 5 \cdot 10^5 = 375000$$

$$0,23 \cdot 2 \cdot 10^5 = 4,25 \cdot 10^3$$

$$\frac{1,7 \cdot 10^6 \text{ BYTE/S}}{1024} = 1660,16 \text{ KB/s}$$

```

SET $t2, $s, $s
BEQ $t2, $0, if
ADD $t0, $t0, $t1
BEQ $s, $s, if
SUB $t0, $t0, $t1

```

```

$5 <- 5      MOVE $5, $0
$6 <- i      MOVE $6, $0
$7 <- n      FOR: BEQ $6, $7, exit
              ADD $5, $5, $6
              ADDI $6, $6, 1
              j FOR
              SW $5, 5($0)

```

```

$5 <- i      WHILE: SET $8, $5, $6
              BEQ $8, $0, ESCI
              ADD $5, $5, $7
              j WHILE
              SW $5, i($0)

```

```

$5 <- i      MULI $6, $6, 2
$6 <- i      WHILE: SET $8, $5, $6
              BEQ $8, $0, ESCI
              ADDI $5, $5, 4
              j WHILE
              EXIT: SW $5, i($0)

```

```

X=0 -> $4   MOVE $5, $0
MAX -> $5    MOVE $6, $0
i=0 -> $6    LW $7, n($0)
n -> $7      FOR: BEQ $6, $7, ESCI
              MULI $8, $6, 4
              ADD $8, $8, $4
              LW $8, X($8)
              ADDI $6, $6, 1
              SET $9, $5, $8
              BEQ $9, $0, ESCI
              MOVE $5, $8
              j FOR
              SW $5, MAX($0)

```

```

$4 -> A[0]   MOVE $8, $0
$8 -> i      SUB $9, $t0, 1
$9 -> j      WHILE: BEQ $8, $t0, ESCI
              MULI $t1, $8, 4
              ADD $t1, $t1, $4
              LW $t1, 0($t1)
              MULI $t2, $9, 4
              ADD $t2, $t2, $4
              LW $t2, 0($t2)
              ADDI $8, $8, 1
              SET $t3, $t1, $t2
              BEQ $t3, $0, ESCI
              MULI $t4, $t2, 2
              SUB $9, $9, 1
              j WHILE
              LW $8, i($8)

```

$\$t = C = 0$

$\$8 \rightarrow i$
 $\$6 \rightarrow n$

```

V($4)      ADDI $29, $29, -8
i($5)      SW $8, 0($29)
j($6)      SW $9, 4($29)
P($7)      MULI $8, $5, 4
              ADD $8, $8, $4
              LW $8, 0($8)
              MULI $9, $6, 4
              ADD $9, $9, $4
              LW $9, 0($9)
              SET $7, $8, $9
              BEQ $7, $0, ESCI
              MOVE $8, $9
              LW $8, 0($29)
              LW $9, 4($29)
              ADDI $29, $29, 8
              SR $31

```

```

V=$4      MOVE $8, $0
W=$5      BEQ $8, $6, ESCI
N=$6      MULI $8, $8, 4
Z(0)=$7   ADD $9, $8, $4
i=$8      LW $5, 0($5)
              ADDI $t0, $8, 5
              LW $t0, 0($t0)
              ADD $t1, $8, $7

```

```

V(0) -> $4  ADDI $29, $29, -16
N -> $5      SW $6, 0($29)
i -> $6      SW $7, 4($29)
n -> $7      SW $9, 8($29)
              SW $31, 12($29)
              MOVE $6, $0
              MULI $7, $5, A-N
              SUBI $5, $5, 1
              BEQ $5, $6, ESCI
              MULI $9, $6, 4
              ADD $4, $4, $5
              SLLI $t0, $t0, 1
              ADDI $6, $6, 1
              SUB $7, $7, 1
              j FOR
              LW $6, 0($29)
              LW $7, 4($29)
              LW $9, 8($29)
              LW $31, 12($29)
              ADDI $29, $29, 16
              jr $31

```

i → \$6	ADDI \$29, \$29, -12	N → \$4	ADDI \$29, \$29, -16	\$11 → B	ADDI \$11, \$12, 100
C → \$5	SW \$7, 0(\$29)		SW \$5, 0(\$29)	\$12 → B	ADDI \$14, \$0, 11
C → \$4	SW \$8, 4(\$29)		SW \$6, 4(\$29)	\$13 → C	MULI \$14, \$14, 4
	SW \$31, 8(\$29)		SW \$7, 8(\$29)	X → 1024	SW \$15, 1024(\$14)
	MUL \$7, \$4, \$4 ^{c²}		SW \$31, 12(\$29)		ADD \$15, \$15, \$13
	MUL \$8, \$5, \$5 ^{c²}		ADDI \$5, \$0, 2		MULI \$14, \$14, 4
	ADD \$7, \$7, \$8		SLT \$6, \$4, \$5 n=2		
	JMC SQRRT		BEQ \$6, \$0, EXSE		
	LW \$7, 0(\$29)		SUBI \$4, \$4, 1		
	LW \$8, 4(\$29)		JBC F4TORACE		
	LW \$31, 8(\$29)		MUL \$7, \$7, \$4		
	ADDI \$29, \$29, 12		LW \$5, 0(\$29)		
	JV \$31		LW \$6, 4(\$29)		
			LW \$7, 8(\$29)		
			LW \$34, 12(\$29)		
			ADDI \$29, \$29, 16		
			JV \$31		

MOVE \$16, \$0
LWI \$16, 60
ADDI \$16, \$16, 8205

f = 100 MHz
32 Bit
3 Cicli di Clock
T = 60 ns
128 Bit

$$T = \frac{1}{f} = \frac{1}{100 \cdot 10^6} = 10 \text{ ns}$$

$$T = 10 \text{ ns} \cdot 3 = 30 \text{ ns}$$

$$3 \cdot \frac{128}{32} \cdot 10 = 120 \text{ ns}$$

$$30 + 120 + 60 = 210 \text{ ns}$$

f = 625 MHz
32 Bit
4 Cicli di Clock
40 ns

$$T = \frac{1}{f} = \frac{1}{625 \cdot 10^6} = 1,6 \text{ ns}$$

$$1,6 \cdot 4 = 6,4 \text{ ns}$$

$$4 \cdot \frac{32}{32} \cdot 1,6 = 6,4 \text{ ns}$$

$$6,4 \cdot 2 + 40 = 52,8 \text{ ns}$$

10⁶ istruzioni/s
6 Clock
32 Bit

65% NO I/O
35% I/O

$$10^6 \cdot 0,35 = 3,5 \cdot 10^5 \text{ istruzioni/s}$$

$$\frac{3,5 \cdot 10^5 \text{ istruzioni/s}}{4 \text{ istruzioni/Parola}} = 87500 \text{ Parole/s}$$

$$87500 \text{ Parole/s} \cdot \frac{32}{8} \text{ Byte} = 350000 \text{ B/s} = 341,78 \text{ KB/s}$$

$$\left. \begin{array}{l} 0,65 \cdot 6 \cdot 10^6 = 3,9 \cdot 10^6 \text{ Parole/s} \\ 0,35 \cdot 2 \cdot 10^6 = 0,7 \cdot 10^6 \text{ Parole/s} \end{array} \right\} 4,6 \cdot 10^6 \text{ Parole/s}$$

$$17 \cdot 968,75 \text{ KB/s}$$

$$0,65 \cdot 2 \cdot 10^5 = 1,3 \cdot 10^5 \text{ Parole/s}$$

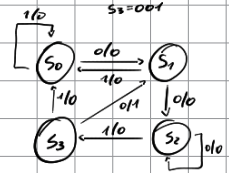
$$0,35 \cdot 5 \cdot 10^5 = 1,75 \cdot 10^5 \text{ Parole/s}$$

$$(1,3 + 1,75) \cdot 10^5 = 3,05 \cdot 10^5 \text{ Parole/s}$$

$$\frac{3,05 \cdot 10^5 \cdot 4}{10^3} = 1220 \text{ KB/sec}$$

0010 $S_1=0$
 $S_2=00$
 $S_3=001$

4 BIT = 4 STATI



STATO ATTUALE	X=0	X=1
S ₀	S ₁ /0	S ₀ /0
S ₁	S ₂ /0	S ₀ /0
S ₂	S ₂ /0	S ₃ /0
S ₃	S ₁ /1	S ₀ /0

$2^2 \cdot 5 \rightarrow 2^2 \cdot 4$

	A	B	X	A'	TA	B'	TB	Z
S_0	0	0	0	0	0	1	1	0
S_1	0	0	1	0	0	0	0	0
S_2	0	1	0	1	1	0	1	0
S_3	0	1	1	0	0	0	1	0
S_0	1	0	0	1	0	0	0	0
S_1	1	0	1	1	0	1	1	0
S_2	1	1	0	0	1	1	0	1
S_3	1	1	1	0	1	0	1	0

X \ AB

	00	01	11	10
0		1	1	
1			1	

$TA = B\bar{X} + AB$

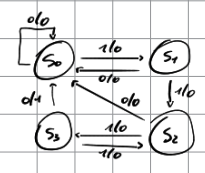
X \ AB

	00	01	11	10
0	1	1		
1		1	1	1

$TB = \bar{A}\bar{X} + BX + AX$

$Z = ABX$

1110 $S_1=1$
 $S_2=11$
 $S_3=111$



STATO ATTUALE	X=0	X=1
S ₀	S ₀ /0	S ₁ /0
S ₁	S ₀ /0	S ₂ /0
S ₂	S ₀ /0	S ₃ /0
S ₃	S ₀ /1	S ₂ /0

$NP = 256 = 2^8 \rightarrow 8 \text{ BIT}$

TA 6 BIT	OFF 2 BIT
----------	-----------

$DIFF = 4 = 2^2 \rightarrow 2 \text{ BIT}$

$\frac{60}{7200} = 8.3 \text{ MS}$

$2^1 2^2 2^3 2^4$ BIT DI CONTROLLO

BIT POSITION	C ₀	C ₁	C ₂	C ₃
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0

- 8 BIT DI INFORMAZIONE $\rightarrow M_1, M_2, \dots, M_8$
- 4 BIT DI CONTROLLO
- LA STRINGA FINALE, BIT DI CONTROLLO + BIT D'INFORMAZIONE, È DI 12 BIT
- 1 BIT DI CONTROLLO STANNO NELLE POSIZIONI $2^1, 2^2, 2^3, 2^4, \dots, 2^8$

1	2	3	4	5	6	7	8	9	10	11	12
C ₀	C ₁	B ₀	C ₂	B ₁	B ₂	B ₃	C ₃	B ₄	B ₅	B ₆	B ₇
1	1	1	1	0	0	0	1	0	0	0	1

DOBBIAMO RITROVARE IL VALORE ASSOCIATO A CIASCUN BIT DI CONTROLLO, DIPENDE DAL CODICE DA RAPPRESENTARE

SE C'È UN NUMERO PARI DI BIT A 1, IL RISULTATO DELLA XOR È ZERO
 SE C'È UN NUMERO DISPARE DI BIT A 1, IL RISULTATO DELLA XOR È 1

SE C'È UN NUMERO PARI DI BIT A 1, IL RISULTATO DELLA XOR È ZERO
 SE C'È UN NUMERO DISPARE DI BIT A 1, IL RISULTATO DELLA XOR È 1

SE C'È UN NUMERO PARI DI BIT A 1, IL RISULTATO DELLA XOR È ZERO
 SE C'È UN NUMERO DISPARE DI BIT A 1, IL RISULTATO DELLA XOR È 1

$C_0 = \text{XOR}(M_1, M_2, M_3, M_4) = 1$

$C_1 = \text{XOR}(M_1, M_2, M_4, M_5) = 1$

$C_2 = \text{XOR}(M_2, M_3, M_6, M_7) = 1$

$C_3 = \text{XOR}(M_3, M_6, M_7, M_8) = 1$

$\frac{1}{f} = 0.5 \text{ NS}$

$0.5 \cdot 2 = 1 \text{ NS}$

$0.5 \cdot 20 = 10 \text{ NS}$

$0.5 \cdot \frac{32}{32} \cdot 2 = 1 \text{ NS}$

12 NS

MIN $2^{-e} \rightarrow 2^{-13}$

MAX $2^{13} \cdot 2^{-13} (2 \cdot 2^{-13}) \rightarrow 2^{-6} (2 \cdot 2^{-6})$

$(45, 25)_{10} = (101101, 01)_2 = 1, 0110101 \cdot 2^5$

45 2	22 2	11 2	5 2	2 2	1 2	0,25 · 2 = 0,5
1 22	0 11	1 5	1 2	0 1	1 0	0,5 · 2 = 1

\$4 \rightarrow A

\$8 \rightarrow i

MUGI \$4, \$4, 4

LW \$4, A (\$4)

ADDI \$4, \$4, 1